NPC — Unit 3 Assignment

Configure a DNS Server and Web Server Using Cisco Packet Tracer

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1. Objective

To build a small LAN in Cisco Packet Tracer with two PCs, a switch, a web server and a DNS server; to enable HTTP on the web server; to map a domain name to the web server's IP address on the DNS server; and to access the website from a PC using the domain name.

2. Topology and IP Plan

All devices connect to one switch (Switch0, 2960) with copper straight-through cables. Network: 192.168.1.0/24.

Device	Type	Interface	IP address	Subnet mask	DNS server
PC0	PC	FastEthernet0	192.168.1.10	255.255.255.0	192.168.1.2
PC1	PC	FastEthernet0	192.168.1.11	255.255.255.0	192.168.1.2
DNS-Server	Server	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.2
Web-Server	Server	FastEthernet0	192.168.1.3	255.255.255.0	192.168.1.2

Domain used: **www.npcsite.com** → 192.168.1.3

Screenshot 1
Network topology (all devices connected, link lights green)

3. Configuration Steps

Step 1 — Build the topology

1. Add 1 switch (2960), 2 PCs and 2 Servers from End Devices.
2. Connect each device's FastEthernet0 to a switch port using a Copper Straight-Through cable.
3. Rename devices (Display Name) as in the table. Wait for the link lights to turn green.

Step 2 — Assign static IP addresses

Click device → **Desktop** → **IP Configuration** → Static, then enter IP, mask and DNS server from the table (for servers it is under **Desktop** → **IP Configuration** too).

Screenshot 2
IP Configuration of PC0 (IP, mask, DNS server)

Screenshot 3
IP Configuration of DNS-Server and Web-Server

Step 3 — Configure the Web Server (HTTP)

1. Web-Server → **Services** → **HTTP**.
2. Make sure **HTTP** is **On** (HTTPS optional).
3. Edit **index.html** (e.g. "Welcome to NPC Web Server — Ruturaj Thorat") and click **Save**.

Screenshot 4
Web-Server → Services → HTTP, with index.html edited and HTTP On

Step 4 — Configure the DNS Server

1. DNS-Server → **Services** → **DNS**; set DNS Service to **On**.
2. Name: **www.npcsite.com**, Type: **A Record**, Address: **192.168.1.3**.
3. Click **Add**, then **Save**.

Screenshot 5
DNS-Server → Services → DNS showing the A record www.npcsite.com → 192.168.1.3

Step 5 — Point the PCs to the DNS server

On PC0 and PC1, set **DNS Server = 192.168.1.2** in IP Configuration (already shown in Screenshot 2).

4. Testing and Results

4.1 Ping by IP (PC0 → Desktop → Command Prompt)

```
C:\> ping 192.168.1.3
C:\> ping 192.168.1.2
C:\> ping 192.168.1.11
```

Screenshot 6
Successful pings (0% loss)

4.2 Ping and DNS lookup by name

```
C:\> ping www.npcsite.com
C:\> nslookup www.npcsite.com
```

Expected: name resolves to 192.168.1.3 and replies are received.

Screenshot 7
ping / nslookup using www.npcsite.com

4.3 Website access

PC0 → **Desktop** → **Web Browser** → enter `http://www.npcsite.com`.

Screenshot 8
Browser showing the web page opened via the domain name

Test	Expected	Result
Ping PC0 → Web server (IP)	Success	Pass / Fail
Ping PC0 → PC1	Success	Pass / Fail
nslookup www.npcsite.com	192.168.1.3	Pass / Fail
Browse http://www.npcsite.com	Page loads	Pass / Fail
Browse from PC1	Page loads	Pass / Fail

5. Troubleshooting Notes

- Browser fails but IP works → DNS service Off, A record missing, or PC DNS address wrong.
- Ping fails → check link lights, IP/mask typos.
- First ping may time out while ARP resolves; later replies succeed.

6. Conclusion

A LAN with a switch, DNS server and web server was built in Cisco Packet Tracer. Static IP addresses were assigned, HTTP was enabled on the web server, and a DNS A record mapped **www.npcsite.com** to 192.168.1.3. Both PCs, configured with the DNS server, resolved the name and opened the website in the browser, showing how DNS lets users reach services by name instead of IP address.